

3N Generator

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February 28, 2025

1 Spectral Function

The 2 nucleon Generalized Contact Formalism (GCF) spectral function is written as follows:

$$S = \frac{1}{4\pi} \sum_{\alpha} C^{\alpha} t^{\alpha} \int \frac{d^3 \vec{p}_2}{(2\pi)^3} \delta(f(\vec{p}_2)) \rho^{\alpha}(\vec{p}_1, \vec{p}_2) n^{\alpha}(|\vec{p}_1 + \vec{p}_2|) \quad (1)$$

Where the basic assumption of contact theory (factorization) manifests itself as the convolution of a 2 body density ($\rho^{\alpha}(\vec{p}_1, \vec{p}_2)$) and the SRC density relative to the nucleus ($n^{\alpha}(\vec{p}_1 + \vec{p}_2)$). The spectral function must also be summed over the quantum numbers of the SRC such as spin and isospin which are both grouped into α . The contacts (C^{α}) effectively weight the probability of each possible SRC state. The delta function is necessary for scattering reactions to conserve energy. I have also included a factor t^{α} which is a combinatorial factor for the number of protons/neutrons in the SRC. For instance, if the electron scatters off of a proton in a pp SRC, then $t^{\alpha} = 2$.

I will generalize the spectral function to a 3N SRC by taking $\rho^{\alpha}(\vec{p}_1, \vec{p}_2) \rightarrow \rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3)$ and $n^{\alpha}(|\vec{p}_1 + \vec{p}_2|) \rightarrow n^{\alpha}(|\vec{p}_1 + \vec{p}_2 + \vec{p}_3|)$. Where the 3N SRC density is generalized to a function of the 3 nucleon momenta and the center of mass distribution is still just sampling from a Gaussian distribution. Finally, we integrate over the third nucleon momentum.

$$S = \frac{1}{4\pi} \sum_{\alpha} C^{\alpha} t^{\alpha} \int \frac{d^3 \vec{p}_2 d^3 \vec{p}_3}{(2\pi)^6} \delta(f(\vec{p}_2, \vec{p}_3)) \rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^{\alpha}(|\vec{p}_1 + \vec{p}_2 + \vec{p}_3|) \quad (2)$$

This is the spectral function we will use for our cross section.

2 SRC Density

The 2 nucleon density functions come from solving the Schrodinger equation for a given potential between 2 nucleons and squaring the wavefunction:

$$\rho^{\alpha}(\vec{p}_1, \vec{p}_2) = |\tilde{\varphi}^{\alpha}(|\vec{p}_1 - \vec{p}_2|)|^2 \quad (3)$$

In the 2 nucleon case, there are 6 variables that can be broken down into 3 center of mass momenta, 2 Euler angles, 1 NN interaction variable. This 1 NN interaction variable is the relative momentum. The 3 nucleon case ($\rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3)$) is much more complicated with 3 center of mass momenta, 3 Euler angles, and 3 NNN interaction variables. So the fundamental NNN interaction is described using only 3 variables. We have calculations for the Helium-3 ground state wavefunction which we will use as a substitute for the 3N SRC wavefunction. The wavefunction is written as:

$$\rho(p_a, p_b, \theta_{a,b}) = |\tilde{\varphi}^{\alpha}(p_a, p_b, \theta_{a,b})|^2 \quad (4)$$

Where p_a is the momentum of particle 1 in the center of mass frame:

$$\vec{p}_a = \vec{p}_1 - \frac{1}{3} \vec{p}_{cm} \quad (5)$$

p_b is the relative momentum between particle 2 and 3:

$$\vec{p}_b = \vec{p}_2 - \vec{p}_3 \quad (6)$$

And the angle $\theta_{a,b}$ is the angle between the 2 vectors. The wavefunction calculation we have is not in the same set of variables as the spectral function currently. Moreover, it only depends upon 3 variables so 6 others will need to be defined.

The variables we will use are as follows. First there is the center of mass motion: $\vec{p}_{cm} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3$. The first two Euler angles are the angles of the vector $\vec{p}_a$ which are $\cos(\theta_a)$ and ϕ_a . The last angle will be a bit more difficult to define. We will want to orient the vectors $\vec{p}_a$ and $\vec{p}_b$ such that $\vec{p}_a \parallel z$. To do this, we rotate $\vec{p}_a$ and $\vec{p}_b$ by $-\phi_a$ about the z axis. Then rotate $\vec{p}_a$ and $\vec{p}_b$ again by $-\theta_a$ about the y axis. Now $\vec{p}_a$ is parallel to z. The two angles that describe $\vec{p}_b$ are $\theta_{b,a||z}$ and $\phi_{b,a||z}$ which describe $\theta_{a,b}$ and the last Euler angle respectively. The density is independent of the center of mass motion and the Euler angles:

$$\rho(p_a, p_b, \theta_{a,b}, \vec{p}_{cm}, \Omega_a, \phi_{b,a||z}) = \rho(p_a, p_b, \theta_{a,b}) \quad (7)$$

Meanwhile, if we want to make this change of variables, we will need to include the Jacobian to switch between them:

$$\rho(p_a, p_b, \theta_{a,b}) = \frac{\partial^3 \vec{p}_1 \partial^3 \vec{p}_2 \partial^3 \vec{p}_3}{\partial^3 \vec{p}_{cm} \partial^2 \Omega_a \partial \phi_{b,a||z} \partial p_a \partial p_b \partial \theta_{a,b}} \rho(\vec{p}_1, \vec{p}_2, \vec{p}_3) \quad (8)$$

Where:

$$\frac{\partial^3 \vec{p}_1 \partial^3 \vec{p}_2 \partial^3 \vec{p}_3}{\partial^3 \vec{p}_{cm} \partial^2 \Omega_a \partial \phi_{b,a||z} \partial p_a \partial p_b \partial \theta_{a,b}} = \frac{1}{p_a^2 p_b^2 \sin(\theta_{a,b})} \quad (9)$$

This Jacobian comes from switching the two vectors to spherical coordinates. We will not need this factor for our calculations.

3 Cross Section

The electron-nucleus cross section we have been using in our simulations factorizes the electron-nucleon part from the nuclear spectral function. It is written:

$$\frac{d^6 \sigma}{d^2 \Omega_k dE_k d^2 \Omega_{p'_1} dE'_1} = p'_1 E'_1 \sigma_{e,N} S \quad (10)$$

E_k is the energy of the scattered electron. p'_1 and E'_1 are the momentum and energy of the struck nucleon. We can make a change of variables to $\vec{p}'_1$ with the appropriate Jacobian.

$$\frac{d^6 \sigma}{d^2 \Omega_k dE_k d^3 \vec{p}'_1} = \sigma_{e,N} S \quad (11)$$

Now plug in the 3N spectral function.

$$\frac{d^6 \sigma}{d^2 \Omega_k dE_k d^3 \vec{p}'_1} = \sigma_{e,N} \frac{1}{4\pi} \sum_{\alpha} C^{\alpha} t^{\alpha} \int \frac{d^3 \vec{p}_2 d^3 \vec{p}_3}{(2\pi)^6} \delta(f(\vec{p}_2, \vec{p}_3)) \rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^{\alpha}(|\vec{p}_1 + \vec{p}_2 + \vec{p}_3|) \quad (12)$$

Change variables again so our cross section will be differential in $\vec{p}_1 = \vec{p}'_1 - \vec{q}$. The Jacobian determinant is 1.

$$\frac{d^6 \sigma}{d^2 \Omega_k dE_k d^3 \vec{p}_1} = \sigma_{e,N} \frac{1}{4\pi} \sum_{\alpha} C^{\alpha} t^{\alpha} \int \frac{d^3 \vec{p}_2 d^3 \vec{p}_3}{(2\pi)^6} \delta(f(\vec{p}_2, \vec{p}_3)) \rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^{\alpha}(p_{cm}) \quad (13)$$

In choosing the variables for the cross section, I decided to keep only 2 electron variables. The variables I have chosen will be the solid angle of the outgoing electron to make my kinematics easier and to more easily generate events in the detector range. We will use E_k to satisfy the delta function. To achieve this, we will integrate both sides by E_k .

$$\frac{d^5 \sigma}{d^2 \Omega_k d^3 \vec{p}_1} = \int dE_k \left[\sigma_{e,N} \frac{1}{4\pi} \sum_{\alpha} C^{\alpha} t^{\alpha} \int \frac{d^3 \vec{p}_2 d^3 \vec{p}_3}{(2\pi)^6} \delta(f(\vec{p}_2, \vec{p}_3)) \rho^{\alpha}(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^{\alpha}(p_{cm}) \right] \quad (14)$$

The delta function can be satisfied by choice of E_k as opposed to $\vec{p}_2$ and $\vec{p}_3$. The conservation of energy relation is:

$$f = m_A + E_{beam} - E_k - \sqrt{m_N^2 + (\vec{p}_1 + \vec{q})^2} - \sqrt{m_N^2 + p_2^2} - \sqrt{m_N^2 + p_3^2} - \sqrt{m_{A-3}^2 + p_{cm}^2} \quad (15)$$

To integrate over E_k we can use the relation: $\delta(f(E_k)) = \delta(E_k - E_k^*)/|\partial f/\partial E_k|$ where E_k^* is the solution to $f(E_k) = 0$. To find $|\partial f/\partial E_k|$, write f as:

$$f = -E_k - \sqrt{m_N^2 + (\vec{p}_1 + \vec{p}_{beam} - E_k \hat{k})^2} + Z \quad (16)$$

Where $\hat{k}$ is the unit vector along the direction of the outgoing electron and $\vec{p}_{beam}$ is the momentum of the incoming electron beam. And:

$$Z = m_A + E_{beam} - \sqrt{m_N^2 + p_2^2} - \sqrt{m_N^2 + p_3^2} - \sqrt{m_{A-3}^2 + p_{cm}^2} \quad (17)$$

The derivative is:

$$|\partial f/\partial E_k| = \frac{E'_1 + \vec{p}_1 \cdot (\vec{p}_1 + \vec{p}_{beam} - \hat{k})}{E'_1} \quad (18)$$

Now our delta function becomes:

$$\delta(f(\vec{p}_2)) = \delta(f(E_k)) = \delta(E_k - E_k^*) \frac{E'_1}{E'_1 + \vec{p}_1 \cdot (\vec{p}_1 + \vec{p}_{beam} - \hat{k})} \quad (19)$$

We can put this back into our cross section and integrate over E_k . From here on, any mention of E_k refers to E_k^* . I will also use J_δ as short hand for the delta function Jacobian until the end.

$$\frac{d^5 \sigma}{d^2 \Omega_k d^3 \vec{p}_1} = \sigma_{e,N} \frac{1}{4\pi} \sum_\alpha C^\alpha t^\alpha \int \frac{d^3 \vec{p}_2 d^3 \vec{p}_3}{(2\pi)^6} \rho^\alpha(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^\alpha(p_{cm}) J_\delta \quad (20)$$

We can remove the 6 integrals by making the cross section differential in $\vec{p}_2$ and $\vec{p}_3$. The sum is also removed by specifying a cross section specific to each set of quantum numbers.

$$\frac{d^{11} \sigma^\alpha}{d^2 \Omega_k d^3 \vec{p}_1 d^3 \vec{p}_2 d^3 \vec{p}_3} = \frac{1}{2(2\pi)^7} \sigma_{e,N} C^\alpha t^\alpha \rho^\alpha(\vec{p}_1, \vec{p}_2, \vec{p}_3) n^\alpha(p_{cm}) J_\delta \quad (21)$$

Now we can make a change of variables. The cross section will change in the following way:

$$\frac{d^{11} \sigma^\alpha}{d^2 \Omega_k d^3 \vec{p}_{cm} d^2 \Omega_a d\phi_{b,a||z} dp_a dp_b d\theta_{a,b}} = \frac{1}{2(2\pi)^7} \sigma_{e,N} C^\alpha t^\alpha \rho^\alpha(p_a, p_b, \theta_{a,b}) n^\alpha(p_{cm}) J_\delta \quad (22)$$

And with the Jacobian substituted in, we have the final cross section:

$$\begin{aligned} \frac{d^{11} \sigma^\alpha}{d^2 \Omega_k d^3 \vec{p}_{cm} d^2 \Omega_a d\phi_{b,a||z} dp_a dp_b d\theta_{a,b}} = \\ \frac{1}{2(2\pi)^7} \sigma_{e,N} C^\alpha t^\alpha \rho^\alpha(p_a, p_b, \theta_{a,b}) n^\alpha(p_{cm}) \frac{E'_1}{E'_1 + \vec{p}_1 \cdot (\vec{p}_1 + \vec{p}_{beam} - \hat{k})} \end{aligned} \quad (23)$$

4 Weight

The weight of a generator is given by $w = \sigma/P$ where P is the probability of sampling a set of variables in your random generation. My choice of generation gives the following value of P :

$$P(\Omega_k, \vec{p}_{cm}, \Omega_a, \phi_{b,a||z}, p_a, p_b, \theta_{a,b}) = \frac{1}{2\pi} \frac{1}{\Delta \cos(\theta_k)} n(\vec{p}_{cm}) \frac{1}{\Delta \cos(\theta_a) \Delta \phi_a \Delta \phi_{b,a||z}} \frac{1}{\Delta p_a \Delta p_b \Delta \theta_{a,b}} \quad (24)$$

With this, we can determine the weight by using the cross section Eq. 23:

$$w = \frac{\Delta \cos(\theta_k)}{2(2\pi)^6} \sigma_{e,N} C^{\alpha} t^{\alpha} \rho^{\alpha}(p_a, p_b, \theta_{a,b}) \frac{E'_1}{E'_1 + \vec{p}_1 \cdot (\vec{p}_1 + \vec{p}_{beam} - \hat{k})} \Delta \cos(\theta_a) \Delta \phi_a \Delta \phi_{b,a||z} \Delta p_a \Delta p_b \Delta \theta_{a,b} \quad (25)$$

5 Kinematics

Even though the generator will be sampling randomly from only 2 electron momentum variables, the last variables (E_K) is fully defined by conservation of energy in the delta function. Eq. 16 gives us the conservation of energy relation:

$$Z - E_k = \sqrt{m_N^2 + (\vec{p}_1 + \vec{p}_{beam} - E_k \hat{k})^2} \quad (26)$$

We can square both sides to solve for E_k , but we will need to ensure that both sides are positive when we randomly sample events.

$$Z^2 + E_k^2 - 2ZE_k = m_N^2 + (\vec{p}_1 + \vec{p}_{beam})^2 + E_k^2 - 2(\vec{p}_1 + \vec{p}_{beam}) \cdot \hat{k} E_k \quad (27)$$

The E_k^2 s drop out and we can solve for E_k by some quick rearranging:

$$E_k = \frac{Z^2 - m_N^2 - (\vec{p}_1 + \vec{p}_{beam})^2}{2(Z - (\vec{p}_1 + \vec{p}_{beam}) \cdot \hat{k})} \quad (28)$$

and ensure that:

$$Z - E_k > 0 \quad (29)$$

If the inequality fails, set weight to zero.

References